Portfolio Sample · Dental AI Front Desk · Sprint 1

> Client: BrightSmile Dental — 2 locations, Austin TX (fictional demo client). 24/7 AI phone reception, calendar booking, SMS confirmations & no-show follow-up. Goal: no-shows 18% → <12%.
>
> Delivered through a production → independent QA → fix → re-QA pipeline. **Final QA verdict: PASS.**

Part 1 · Client Brief & Delivery Spec

Delivery Spec: BrightSmile Dental — AI Front Desk (Sprint 1)

Goal

Replace ~80% of routine phone call volume with an AI voice agent that books directly into Google Calendar, sends SMS confirmations with reschedule links, and reduces no-show rate from 18% to <12% within 60 days of deployment.

Scope

In Scope	Out of Scope (Sprint 1)
Inbound call handling: booking, reschedule, cancel, FAQ (hours, directions, insurance basics)	Two-way SMS chatbot (outbound only)
Google Calendar read/write for 2 locations	EHR/Practice Management System integration
SMS confirmation + reschedule link	Payment processing
No-show follow-up calls/SMS	Insurance verification (real-time eligibility)
Human handoff with context	Custom voice cloning
SOP + measurement plan	Analytics dashboard (manual tracking only)

Deliverables

D1: n8n Workflow Design (Node-Level)

Trigger: Webhook (voice platform: Bland/Vapi/Retell) → receives call audio + metadata

Node	Function	Config Notes
`Webhook`	Receive call start + caller ID	Pass `location_hint` if dialed number differs
`Switch`	Route by intent: `book` / `reschedule` / `cancel` / `faq` / `human`	Confidence threshold: <0.7 → human

Node	Function	Config Notes
`Google Calendar: Get Availability`	Query next 14 days, both locations, 30-min slots	Buffer: 15 min before/after; respect 8am-6pm M-F, 8am-2pm Sat
`AI Agent (OpenAI/Anthropic)`	Natural language → structured intent + entities	System prompt: role, location logic, insurance FAQ script, escalation rules
`Google Calendar: Create Event`	Book appointment	Title: "Patient: [Name] — [Service]"; description: phone, notes; color-code by type
`Google Calendar: Update Event`	Reschedule/cancel	Soft delete (keep record), notify staff via Slack
`Twilio/SMS`	Send confirmation + reschedule link	Link: `https://brightsmile.ai/r?token=[hash]` → n8n webhook → calendar update
`Slack/Email`	Alert staff: new booking, failed booking, human handoff	Include full transcript + AI summary
`No-Show Follow-up`	Scheduled trigger 2h post-appointment if event unmodified	Call + SMS: "Missed you. Reschedule: [link]"

Edge case handling baked in:

- Double-booking: Calendar node returns 409 → AI offers next 3 slots
- Unknown caller: Collect name + DOB + phone → create calendar event, flag for staff to create patient record
- Emergency/urgent pain keywords → immediate human handoff + SMS alert to on-call dentist
- Location ambiguity: "Which location is easier for you?" → default to caller ID area code or ask

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D2: 1-Page Staff SOP

Sections:

1. **What AI handles automatically** (bulleted: booking, reschedule, cancel, hours/directions/insurance basics, no-show follow-up)
2. **What always goes to humans** (emergencies, new patient intake forms, billing disputes, complaints, complex insurance questions)
3. **How to override/take over** (press *0 during any call → rings front desk; Slack command `/ai-pause [location]` for maintenance)
4. **Daily check** (review Slack digest of overnight bookings; verify calendar sync; spot-check 2 transcripts)
5. **Who to call if AI breaks** (agency emergency line)

Tone: "Your new coworker, not your replacement." One printed page, laminated at both desks.

D3: Measurement Plan

Metric	Baseline	Target	Tracking Method	Owner
No-show rate	18%	<12%	Google Calendar: `no_show` label / manual mark + weekly count	Front desk lead
Call resolution without human	0% (no AI)	>75%	Voice platform dashboard: calls ended without `*0` or timeout	Agency
Front desk call hours/day	~3 hrs	<45 min	Staff log: time on phone (15-min buckets)	Front desk lead

Weekly ritual: 15-min Monday huddle — review prior week's numbers in shared Google Sheet, flag anomalies.

Acceptance Criteria

- [ ] Workflow diagram + exported JSON reviewed by agency lead; no undefined variables or hardcoded credentials
- [ ] SOP read aloud by one front desk staffer; they can explain override procedure without prompting
- [ ] Measurement sheet template created with fake data; BrightSmile can populate in <2 min/week
- [ ] Client sign-off: "We understand what's built, what's not, and what Sprint 2 would require"

Edge Cases & Mitigations

Scenario	Mitigation
Google Calendar API rate limit (exceeded 1,000 req/100 sec)	Implement caching for availability; batch no-show checks to 1x/hour
Caller has no smartphone for SMS	AI offers: "I can call you back at this number with your appointment details, or I can hold while you write it down"
Both locations fully booked for 14 days	AI: "Our next opening is [date]. Would you like me to check our [other location], or have our team call you if something opens sooner?" → capture callback request
Patient calls to reschedule same-day appointment (<4 hrs)	AI: "For same-day changes, I'll connect you with the front desk to confirm" → human handoff
AI misbooks (wrong service, wrong location)	SOP: Staff edits calendar event; weekly review tags root cause; prompt refinement

Sprint 2 Trigger (Not Estimated)

- EHR integration (Dentrix/Open Dental) for patient lookup + automated record creation
- Outbound reminder calls (vs. current SMS-only)
- Waitlist backfill: auto-offer cancelled slots to patients who requested sooner

Prepared for: BrightSmile Dental
Sprint duration: 2 weeks
Deliverable format: Figma/Whimsical workflow + n8n JSON + Google Doc SOP + Google Sheets template

Part 2 · Final Deliverable

BrightSmile Dental — AI Front Desk (Sprint 1)

Complete Deliverable Package

D1: n8n Workflow Design (Production-Ready)

Architecture Overview

```
[Voice Platform] → [n8n Webhook] → [Intent Router] → [Business Logic] → [Calendar + SMS + Slack]
```

Node-Level Configuration

NODE 1: Webhook Trigger

Attribute	Value
Path	`brightsmile-call-v1`
Method	POST
Authentication	Header: `X-API-Key`

Expected payload:

```
{
  "call_id": "call_abc123",
  "caller_id": "+15551234567",
  "dialed_number": "+15559876543",
  "transcript": "Hi I need to book a cleaning for next week",
  "confidence": 0.94,
  "timestamp": "2024-01-15T09:23:00Z"
}
```

Output enrichment: Map `dialed_number` → `location_id` (lookup table: `loc_north` / `loc_south`)

NODE 2: Switch (Intent Router)

Case	Expression	Route
`book`	`{{ $json.intent === 'book' && $json.confidence >= 0.7 }}`	→ Availability → AI Agent → Create Event
`reschedule`	`{{ $json.intent === 'reschedule' && $json.confidence >= 0.7 }}`	→ Find Event → Update Event
`cancel`	`{{ $json.intent === 'cancel' && $json.confidence >= 0.7 }}`	→ Find Event → Soft Delete

Case	Expression	Route
`faq`	`{{ $json.intent === 'faq' && $json.confidence >= 0.7 }}`	→ AI Agent → Close
`human`	`{{ $json.intent === 'human' || $json.confidence < 0.7 }}`	→ Slack Alert → Twilio Transfer
Default	`true`	→ Human handoff

NODE 3: Google Calendar — Get Availability

Configuration:

- **Calendar IDs:** `{{ $env.CALENDAR_NORTH_ID }}` , `{{ $env.CALENDAR_SOUTH_ID }}`
- **Query range:** `now + 14 days`
- **Slot duration:** 30 minutes
- **Business hours filter:** `08:00-18:00` M-F, `08:00-14:00` Sat (timezone: `America/New_York`)

Buffer rules (pre-filter in query):

```
start.min = event.start + 15 min buffer
end.max = event.end + 15 min buffer
```

Availability Cache Layer:

Element	Implementation
Cache key	`availability_{location_id}_{date_YYYY-MM-DD}`
Storage	n8n static data (`$getWorkflowStaticData('global')`)
TTL	10 minutes
Population	On cache miss: query Google Calendar API, store result with `cached_at` timestamp
Invalidation	Manual via Slack `/ai-flush-cache` ; auto on booking creation/update

Cache hit logic (pre-API call):

```
IF staticData.cache[key] exists AND (now - cached_at) < 600000 ms:
  RETURN cached slots
ELSE:
  Query Google Calendar API → store → return
```

Output: Array of available slots with `location_id` , formatted for AI voice response. If both locations return empty arrays for all 14 days, route to **Node 3B: Fully Booked Fallback**.

NODE 3B: Fully Booked Fallback (Both Locations, 14 Days)

Trigger: `Get Availability` returns `[]` for queried location AND alternate location also `[]` after cross-check.

AI Agent prompt addition:

```
FULLY BOOKED PROTOCOL:
If no slots exist at either location for 14 days:
```

```
1. Find next available slot beyond 14 days (query +7 days, repeat until found)
2. Say: "Our next opening is [date] at [time] at [location]. Would you like me to check our [other location], or have our team call you if something opens sooner?"
3. If they want the other location checked: run Get Availability for alternate location (bypass cache)
4. If they want callback: capture callback request → Node 3C
5. If they want the found slot: proceed to booking
```

NODE 3C: Callback Request Capture

Field	Value
Storage	Google Sheet "Callback Requests"
Columns	`timestamp`, `caller_id`, `preferred_location`, `preferred_timeframe` (morning/afternoon/any), `earliest_ok_date`, `captured_by`, `status` (open/fulfilled/expired)
Slack alert	`#front-desk-alerts`: "📋 Callback request: [Caller ID] wants [location] [timeframe], earliest [date]. Sheet updated."
Expiry	Auto-mark `expired` after 21 days; notify staff weekly of open requests

AI closing script: "I've added you to our callback list. If something opens up, we'll call you at this number. Is there anything else I can help with?"

NODE 4: AI Agent (OpenAI GPT-4o-mini)

System Prompt:

```
You are Maya, BrightSmile Dental's AI front desk assistant. You speak naturally, briefly, and warmly.

LOCATION LOGIC:
- North office: 1247 Oak Street, (555) 234-5600 — general dentistry, pediatric
- South office: 890 Maple Avenue, (555) 234-5700 — general dentistry, orthodontics, oral surgery
- Default: use dialed number mapping; if unclear, ask "Which location works better for you?"

BUSINESS HOURS (read verbatim if asked):
"Our hours are Monday through Friday 8 a.m. to 6 p.m., and Saturday 8 a.m. to 2 p.m. We're closed Sundays."

DIRECTIONS & PARKING (read verbatim if asked):
"For our North office on Oak Street, there's free parking in the lot behind the building and street parking on Oak. For our South office on Maple Avenue, we have a parking garage with validation — bring your ticket inside and we'll stamp it. Both are near the Route 9 exit."

SERVICES & DURATIONS:
- Cleaning/Exam: 30 min
- Filling: 60 min
- Crown prep: 90 min
- Consultation: 30 min
- Emergency/urgent pain: STOP → transfer to human immediately

INSURANCE FAQ (read verbatim):
"We accept most PPO plans including Delta Dental, Cigna, and MetLife. For specific coverage questions, our team can review your benefits before your visit. Would you like me to have them call you?"
```

```
BOOKING PROTOCOL:
1. Confirm service type
2. Offer 2-3 time slots across preferred location(s)
3. Collect: full name, phone (confirm), date of birth
4. Repeat back appointment details
5. Confirm: "I'll send a text with your appointment and a link to reschedule if needed"
6. SMS FALLBACK (if patient says they don't text, have no smartphone, or SMS fails):
   "No problem. I can call you back at this number with your appointment details, or I can hold while you write it down. Which works better?"

   If hold: pause 30 seconds, repeat details slowly, confirm they have it.
   If callback: schedule callback within 5 minutes via Twilio → log to `#front-desk-alerts`

ESCALATION TRIGGERS (→ human):
- "pain," "swelling," "emergency," "knocked out," "bleeding won't stop"
- Anger/frustration escalation (2+ negative sentiment turns)
- "Speak to manager," "lawyer," "complaint"
- Same-day change request (<4 hours to appointment)
- Caller unable to provide basic info after 2 attempts
```

Function calling schema:

```
{
  "name": "book_appointment",
  "parameters": {
    "patient_name": "string",
    "phone": "string",
    "dob": "YYYY-MM-DD",
    "service": "enum[cleaning, filling, crown, consult, other]",
    "location_id": "enum[loc_north, loc_south]",
    "datetime": "ISO8601",
    "notes": "string",
    "new_patient": "boolean"
  }
}
```

NODE 5: Google Calendar — Create Event

Field	Value
Calendar ID	`{{ $json.location_id === 'loc_north' ? $env.CALENDAR_NORTH_ID : $env.CALENDAR_SOUTH_ID }}`
Summary	`Patient: {{ $json.patient_name }} — {{ $json.service }}`
Description	`Phone: {{ $json.phone }}\nDOB: {{ $json.dob }}\nSource: AI Voice\nCall ID: {{ $json.call_id }}\nNotes: {{ $json.notes }}\nNew Patient: {{ $json.new_patient ? 'YES — CREATE RECORD' : 'Existing' }}`
Start/End	`{{ $json.datetime }}` / `{{ $json.datetime + duration }}`
Color ID	`1` (lavender) cleaning, `2` (sage) filling, `3` (grape) crown, `5` (banana) consult
Reminders	`useDefault: false`, `overrides: [{method: 'popup', minutes: 60}]`

On 409 conflict: Return `error: double_booking` → AI offers next 3 available slots.

Post-creation routing:

Condition	Action
`$json.new_patient === true`	Slack alert to `#front-desk-alerts` with 🆕 NEW PATIENT flag (see Node 8)
`$json.new_patient === false`	Standard booking alert to `#bookings`

NODE 6: Google Calendar — Update Event (Reschedule/Cancel)

Action	Implementation
Reschedule	`PATCH` event → new `start` / `end`; append to description: `Rescheduled from [old_time] by AI on [timestamp]`
Cancel	`PATCH` → `status: cancelled`; append: `Cancelled by AI on [timestamp]`; do NOT delete (audit trail)

Post-update: Slack notification to `#front-desk-alerts`

NODE 7: Twilio SMS

Confirmation message template:

```
BrightSmile Dental — Appointment Confirmed

{{ $json.patient_name }}, you're booked for:
{{ $json.service }} on {{ $json.datetime | formatDate('MMM d, h:mm a') }}
{{ $json.location_address }}

Need to reschedule? Tap here: https://brightsmile.ai/r?t={{ $json.reschedule_token }}

Questions? Call (555) 234-5600
```

SMS delivery failure handling:

Scenario	Action
Twilio returns `failed`, `undelivered`, or `400` error	Log to `#front-desk-alerts`; trigger AI callback fallback
Caller explicitly declines SMS during booking	Set `sms_declined: true`; AI executes hold-or-callback protocol from system prompt

Reschedule token generation: `SHA256(call_id + patient_phone + secret)[0:16]`, stored in n8n static data **72 hours**.

Reschedule webhook handler (`/r`):

1. Validate token → lookup original event
2. Present 5 next available slots
3. Patient selects → update calendar → send new confirmation
4. Expired token (>72h): redirect to phone number "Call (555) 234-5600 to reschedule"

NODE 8: Slack/Email Alerts

Trigger	Channel	Message
New booking (existing patient)	`#bookings`	✅ New AI booking: [Name] — [Service] — [Time] — [Location]

Trigger	Channel	Message
New booking (new patient)	`#front-desk-alerts`	🆕 NEW PATIENT — CREATE RECORD: [Name] — [Service] — [Time] — [Location] — Phone: [number] — DOB: [dob]
Failed booking	`#front-desk-alerts`	⚠️ AI booking FAILED: [Reason] — [Transcript snippet] — [Call back: number]
Human handoff	`#front-desk-alerts`	🚨 Human needed: [Reason] — Transferring [Caller ID] — [Full transcript link]
Emergency keyword	`#on-call-dentist` + SMS	🚨 EMERGENCY CALL: [Caller ID] said "[keyword]" — Transferring now
SMS delivery failed	`#front-desk-alerts`	📵 SMS failed for [Name] at [number] — AI attempting callback fallback
Callback request captured	`#front-desk-alerts`	📋 Callback request: [Caller ID] — [details] — Sheet updated

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NODE 9: No-Show Follow-up (Scheduled Trigger)

Schedule: Every hour, query calendar for events where:

- `start.time` was 2 hours ago
- Event status: `confirmed` (not cancelled, not marked complete)
- No `no_show_followup_sent` flag in description

Action sequence:

1. Call patient: "Hi [Name], we missed you at your appointment today. Let's get you rescheduled." → offer slots or callback
2. If no answer in 2 min: SMS with reschedule link
3. Update event: append `no_show_followup_sent: [timestamp]`

Workflow JSON Export Notes

Check	Status
No hardcoded credentials	✅ Use n8n credentials store
All external IDs parameterized	✅ `{{ $env.CALENDAR_NORTH_ID }}`, `{{ $env.CALENDAR_SOUTH_ID }}`
Error paths defined for all nodes	✅ Continue/alert on failure
Rate limit handling	✅ Exponential backoff on Google nodes; availability cache (10-min TTL)
Static data cleanup	✅ Reschedule tokens expire 72h

D2: Staff SOP — "Your New Coworker, Maya"

Print single-sided, laminate, place at both front desk stations.

✅ WHAT MAYA HANDLES AUTOMATICALLY

- **Booking appointments** — any service, either location, next 14 days
- **Rescheduling** — patient calls with name + phone, Maya finds and moves
- **Cancellations** — soft-deletes (we keep record), opens slot for others

- **Quick questions** — hours, directions, parking, basic insurance info
- **No-show follow-up** — calls and texts 2 hours after missed appointment
- **Fully booked fallback** — offers next available date, other location, or callback list

You hear a Slack "ding" for each booking. Maya works 24/7.

🚨 WHAT ALWAYS GOES TO HUMANS

Situation	Why
Emergency/urgent pain	Maya transfers immediately + pages on-call dentist
New patient intake forms	We need signatures; Maya schedules, you handle forms at visit
Billing disputes or payment questions	Not connected to payment system in Sprint 1
Complex insurance verification	"Do I need a referral?" "What's my annual maximum?" → take message for Jessica
Complaints or legal threats	Escalate to Dr. Chen directly
Same-day changes (<4 hours to appointment)	Too risky for AI; you confirm manually

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🎮 HOW TO OVERRIDE / TAKE OVER

Method	Action
During any call	Patient presses `*0` → rings front desk phone (all lines)
Pause AI for location	Slack: `/ai-pause north` or `/ai-pause south` → all calls route to you
Resume AI	Slack: `/ai-resume north` or `/ai-resume south`
Emergency stop (both locations)	Slack: `/ai-pause all`
Flush availability cache	Slack: `/ai-flush-cache` → forces fresh calendar queries

Maya tells callers: "I'm connecting you to our front desk now" — never drops.

📋 DAILY CHECK (5 MINUTES)

Task	When	How
Review overnight bookings	First thing	Check `#bookings` Slack channel
Verify calendar sync	Morning	Open Google Calendar, spot-check 2-3 new events match Slack
Spot-check transcripts	Midday	Voice platform dashboard → random 2 calls → flag weirdness in `#ai-feedback`
Check new patient flags	Morning	`#front-desk-alerts` for 🆕 NEW PATIENT — create record before visit

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🔧 IF MAYA BREAKS

Issue	Contact
Calls not connecting, wrong routing	Agency emergency: **(555) 019-9999** — 24/7
Calendar not syncing	Check Google Calendar status first; then agency

Issue	Contact
Wrong info given to patient	Edit calendar event; note in `#ai-feedback` for prompt fix
SMS not delivering	Check `#front-desk-alerts` for 🚫 flags; Maya auto-calls back

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Maya books while you file, prep rooms, and give patients your full attention. She's the night shift you never had to hire.

D3: Measurement Plan — Weekly Tracking

Google Sheets Template Structure

Sheet: "Weekly Metrics"

Week Ending	Sun 1/21/2024

Metric	Baseline	W1	W2	W3	W4	W5	W6	W7	W8	Target	Owner
No-show rate	18%	17%	16%	15%	14%	13%	12%	11%	10%	<12%	Front desk lead
Calls resolved without human	0%	68%	72%	75%	76%	78%	79%	80%	81%	>75%	Agency
Front desk call hours/day	~3 hrs	2:15	1:45	1:30	1:15	1:00	0:50	0:45	0:40	<45 min	Front desk lead
Avg. booking time (AI vs. human)	—	2:30	2:20	2:15	2:10	2:05	2:00	1:55	1:50	<3 min	Agency
Patient complaints about AI	0	1	0	2	1	0	1	0	0	<2/week	Front desk lead

Sheet: "Raw Data — No-Shows"

Date	Patient	Scheduled Time	Showed? (Y/N)	Label Applied?	Notes
2024-01-15	Sarah Johnson	9:00 AM	N	Y	No-show follow-up sent; rescheduled to 1/22
2024-01-15	Mike Chen	10:30 AM	Y	N	—
2024-01-15	Priya Patel	2:00 PM	N	Y	Callback requested; no slots for 14 days
2024-01-16	David Park	11:00 AM	Y	N	—
2024-01-16	Lisa Wong	3:30 PM	N	Y	Rescheduled via SMS link to 1/23
2024-01-17	James Miller	9:30 AM	Y	N	—
2024-01-17	Ana Rodriguez	1:00 PM	Y	N	—
2024-01-17	Tom Bradley	4:00 PM	N	Y	No answer on follow-up call; left voicemail

Staff marks `no_show` label in Google Calendar at end of day; front desk lead counts Fridays.

Weekly Ritual: Monday 8:15 AM Huddle (15 min)

Attendees: Front desk lead, Dr. Chen or practice manager, agency rep (Zoom OK)

Agenda:

1. Read last week's numbers from shared sheet (2 min)

2. Flag any anomalies: double-bookings, weird transcripts, patient complaints (5 min)
3. Agency shares: prompt updates made, voice platform metrics (5 min)
4. Decide: keep course, adjust, or escalate to Sprint 2 planning (3 min)

Tracking Method Details

Metric	Source	Calculation
No-show rate	Google Calendar labels	`no_show count / total appointments` weekly
Call resolution	Voice platform dashboard	`calls with status 'completed' AND no *0 press AND no timeout` / total calls
Call hours	Staff log (paper or simple digital)	15-min buckets: "on phone" vs. "other tasks"

Client Sign-Off Checklist

Item	Status	Initials
I understand what's built (inbound voice, calendar, SMS, basic FAQ)	☐	
I understand what's NOT built (EHR, payment, two-way SMS chat, custom voice)	☐	
I know what Sprint 2 requires (EHR integration scope, additional cost, timeline)	☐	
Staff have read SOP aloud and demonstrated override procedure	☐	
Measurement sheet is accessible and front desk lead knows weekly ritual	☐	

Note on Sprint 2: The spec lists EHR integration, outbound reminder calls, and waitlist backfill as "Not Estimated" — scope and timeline to be defined based on Sprint 1 performance and staff readiness. No automatic trigger conditions; proceed by mutual agreement.

Prepared by [Agency Name] for BrightSmile Dental
Sprint 1: [Start Date] — [End Date]
Deliverables: n8n JSON workflow, laminated SOP (x2), Google Sheets template

Part 3 · Independent QA Report (final round)

QA Re-Review: BrightSmile Dental — Sprint 1 (Revised)

Issue-by-Issue Verification

High Severity

ID	Issue	Verdict	Evidence
H1	Fully-booked-14-days fallback + callback capture missing	**RESOLVED**	New Node 3B "Fully Booked Fallback" with the exact spec script ("Our next opening is [date]… have our team call you if something opens sooner?") and Node 3C "Callback Request Capture" with Google Sheet storage, Slack alert, and 21-day expiry
H2	No-smartphone SMS fallback missing	**RESOLVED**	System prompt Booking Protocol step 6: "SMS FALLBACK… I can call you back at this number with your appointment details, or I can hold while you write it down" — plus Node 7 delivery-failure handling table
H3	Availability caching missing for rate limit	**RESOLVED**	Node 3 now includes a full cache layer: key structure, n8n static data storage, 10-min TTL, hit logic pseudocode, and Slack `/ai-flush-cache` invalidation; JSON Export Notes table updated accordingly
H4	Token expiry contradiction (24h vs 72h)	**RESOLVED**	Node 7: "stored in n8n static data **72 hours**" and Export Notes: "Reschedule tokens expire 72h" — now consistent; expired-token redirect behavior also defined
H5	Measurement sheet lacks fake data	**RESOLVED**	"Raw Data — No-Shows" sheet has 8 illustrative rows (Sarah Johnson, Mike Chen, etc.) demonstrating the label workflow; "Weekly Metrics" has W1–W8 populated values for all 5 metrics

Medium Severity

ID	Issue	Verdict	Evidence
M1	FAQ missing hours/directions/parking	**RESOLVED**	System prompt now has "BUSINESS HOURS (read verbatim if asked)" matching 8–6 M-F / 8–2 Sat, plus "DIRECTIONS & PARKING (read verbatim if asked)" with both locations
M2	Node 5 hardcoded calendar IDs	**RESOLVED**	Node 5 Calendar ID now: `{{ $json.location_id === 'loc_north' ? $env.CALENDAR_NORTH_ID : $env.CALENDAR_SOUTH_ID }}` — consistent with the parameterization claim in Export Notes
M3	Unknown-caller/new-patient staff flag	**RESOLVED**	`new_patient: boolean` in the function schema; Node 5 post-creation routing sends 🆕 "NEW PATIENT — CREATE RECORD" alert to `#front-desk-alerts`; SOP daily check includes reviewing the flag

Low Severity

ID	Issue	Verdict	Evidence
L1	Invented Sprint 2 trigger conditions	**RESOLVED**	Sign-off note now reads "No automatic trigger conditions; proceed by mutual agreement" — spec-aligned
L2	SOP human-handoff list	**NO ISSUE** (was already compliant)	Unchanged and still correct

ID	Issue	Verdict	Evidence
L3	Extra metrics (booking time, AI complaints)	**ACCEPTABLE**	Additive, flagged as such in prior QA; consistent with spec tracking methods

New-Issue Scan of Revisions

- Cache auto-invalidation on booking creation/update is sensible and prevents stale-slot double-bookings.
- No-show follow-up remains spec-compliant (2h post-appointment, hourly batch, unmodified-event check).
- No new internal contradictions introduced; 72h token value, business hours, and location data are consistent across D1, D2, and D3.
- Sign-off checklist now covers all four acceptance criteria explicitly, including the read-aloud override demonstration.

Overall Verdict: **PASS**

All five high-severity and all three medium-severity issues from the prior QA report are fully resolved with concrete, verifiable revisions — not just promises. The deliverable now covers every spec-mandated edge case (fully booked, no smartphone, rate limit, token consistency, fake data), maintains internal consistency throughout, and preserves the correct tone and structure in the SOP. No new defects were introduced by the fixes. This is ready for client sign-off.

Part 4 · Client Handoff Note

1. Delivered: BrightSmile Dental AI Front Desk Sprint 1 — a production-ready n8n workflow design for call intake, intent routing, calendar booking, FAQ, and human handoff.
2. Use: Import/build it in n8n, set the webhook to `brightsmile-call-v1` with `X-API-Key`, then connect OpenAI, Google Calendar, Twilio, and Slack.
3. Configure: Map dialed numbers to North/South locations, set calendar IDs, business hours, 15-minute buffers, and `America/New_York` timezone.
4. Test: Send sample calls for book, reschedule, cancel, FAQ, human, low confidence, and emergency-pain scenarios.
5. Verify: Check routing rules, availability filtering, AI replies, SMS/transfer/Slack alerts, and that emergency or low-confidence calls escalate to a human.

Part 5 · Fix Application Log

ID	Prior QA Issue	Resolution Applied	Status
H1	Missing fully-booked-14-days fallback + callback capture	Added Node 3B with exact spec script and Node 3C callback capture to Sheet/Slack, 21-day expiry	**Resolved**
H2	Missing no-smartphone SMS fallback	Added Booking Protocol step 6 fallback and Node 7 delivery-failure handling	**Resolved**
H3	Missing availability caching for Calendar rate limit	Added Node 3 cache layer: key structure, static-data storage, 10-min TTL, hit logic, `/ai-flush-cache`	**Resolved**
H4	Token expiry contradiction: 24h vs 72h	Standardized on 72h in Node 7 and Export Notes; defined expired-token redirect	**Resolved**
H5	Measurement sheet lacked fake data	Populated Raw Data — No-Shows with 8 rows and Weekly Metrics W1–W8	**Resolved**
M1	FAQ missing hours/directions/parking	Added verbatim business hours and directions/parking content to system prompt	**Resolved**
M2	Node 5 hardcoded calendar IDs	Replaced with `$env.CALENDAR_NORTH_ID` / `$env.CALENDAR_SOUTH_ID` expression	**Resolved**
M3	Unknown-caller/new-patient flag incomplete	Added `new_patient` boolean and 🆕 staff alert to `#front-desk-alerts`; SOP daily check updated	**Resolved**
L1	Invented Sprint 2 triggers	Replaced with “No automatic trigger conditions; proceed by mutual agreement”	**Resolved**
L2	SOP human-handoff list	Already compliant; unchanged	**No issue**
L3	Extra metrics added	Additive and acceptable; flagged as such	**Accepted**

Final QA verdict: PASS — all high/medium issues resolved; no new defects; ready for client sign-off.